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Image-to-Biomarker Prediction of CSF Amyloid- β via Multimodal Deep Learning on MRI and Brain Perfusion SPECT

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Abstract

Alzheimer's disease (AD) is the most common form of dementia. Recently, anti-amyloid- β (A β) antibody therapy has been applied to treat early AD. Therefore, the diagnosis of early AD has become more important than ever before. Cerebrospinal fluid (CSF) biomarkers such as amyloid- β (A β) and tau proteins are widely recognized indicators of AD pathology. Lumbar puncture to obtain CSF is invasive, and this procedure tends to be unfavorable for patients and their families. Although amyloid positron emission tomography (PET) is another method to confirm AD pathology, it is expensive. Single photon emission computed tomography (SPECT), which shows regional cerebral blood flow, has the advantage of being relatively inexpensive and can be used for repeated examination under medical insurance coverage in dementia clinical practice. To address this limitation, we present an image-to-biomarker deep learning framework for non-invasive prediction of CSF A β levels by leveraging multimodal neuroimaging. Our method integrates structural information from T1-weighted MRI and functional information from brain perfusion SPECT. All images are spatially normalized using DARTEL and processed using a ResNet-based 3D convolutional neural network (CNN). A multitask regression model is employed to simultaneously predict A β 42, A β 40, and the A β 42/40 ratio, which is clinically valued for its robustness to individual variability, with age incorporated as an auxiliary input to improve prediction performance. The model is trained using a weighted mean squared error loss function and evaluated through five-fold cross-validation on a dataset of 153 patients with dementia. Ablation studies demonstrate that including both brain perfusion SPECT and age information significantly improves prediction performance, achieving root mean squared errors

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(RMSEs) of 500 pg/mL for A β 42, 3,838 pg/mL for A β 40, and 0.0355 for the A β 42/A β 40 ratio. These findings underscore the potential of our image-to-biomarker approach as a practical and accurate solution for CSF biomarker prediction, offering promising applications in improving the accuracy of early AD diagnosis, clinical decision-making support, and reducing the psychological, physical, and financial burden on patients and their families.

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1. Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder and the leading cause of dementia worldwide. In Japan, the number of elderly individuals living with dementia is expected to exceed 6.45 million by 2060, underscoring the urgent need for effective early detection strategies.

Immunotherapies such as monoclonal antibodies targeting amyloid- β (A β) have been approved in recent years, but their therapeutic effect is limited to slowing disease progression during the early stages, including the prodromal phase [1, 2]. Consequently, detecting AD before irreversible neuronal damage occurs remains a significant challenge in clinical practice.

Identifying AD pathology is essential for initiating treatment during the early stages of disease. Currently, there are two ways to confirm AD pathology: one is to measure CSF biomarkers, and the other is to perform amyloid positron emission tomography (PET) imaging.

Among the biological indicators of AD pathology, cerebrospinal fluid (CSF) biomarkers (including A β 42, A β 40, and A β 42/A β 40) are well recognized as reliable tools [3–5]. However, lumbar puncture, the standard procedure for CSF collection, is invasive and carries risks such as post-procedural headache or infection [6, 7], which limits its applicability in routine clinical settings.

As non-invasive alternatives, structural magnetic resonance imaging (MRI) can detect characteristic patterns of brain atrophy, and is routinely acquired as part of standard diagnostic workflows for dementia. This makes MRI a widely available and clinically established modality for computational biomarker estimation. However, MRI primarily captures anatomical changes and may not fully reflect the functional alterations associated with AD pathology.

Amyloid PET is another method to confirm AD pathology; however, it is expensive and typically covered for only a single scan under medical insurance. Single photon emission computed tomography (SPECT), which provides information on regional cerebral blood flow patterns [8], offers the advantage of being relatively inexpensive and can be performed repeatedly under insurance coverage in dementia clinical practice. Brain perfusion SPECT is also used clinically, particularly to differentiate subtypes of neurodegenerative dementia.

While previous research has primarily utilized MRI alone for imaging-based prediction tasks [9–11], particularly for classifying amyloid PET status, few studies have attempted to estimate continuous CSF biomarker levels, and fewer have integrated MRI

and brain perfusion SPECT in a unified framework. SPECT offers complementary physiological information that may provide additional value to predictive modeling by capturing functional brain alterations. In particular, it remains unclear whether integrating functional SPECT with structural MRI can improve the accuracy of CSF biomarker prediction beyond using MRI alone.

In this study, we address this question by proposing a deep learning framework for non-invasive prediction of CSF A β 42, A β 40, and the A β 42/A β 40 ratio using both T1-weighted MRI and brain perfusion SPECT. Given that MRI is already part of routine clinical assessment, we do not consider SPECT-only scenarios, as they lack structural context and are less reflective of real-world practice. Our model employs a 3-D ResNet-based multitask regression architecture and incorporates demographic variables to improve prediction accuracy. Cross-validation and ablation analyses demonstrate the benefit of this multimodal image-to-biomarker approach as a practical and scalable alternative to lumbar puncture for estimating AD-related biomarkers.

2. Related Works

Previous studies have investigated machine learning approaches to predict amyloid pathology using MRI as the sole imaging modality. Momota et al. (2024) [9] used source-based morphometry with support vector machines to classify amyloid PET positivity across multiple neurological conditions. Their model achieved high accuracy but relied solely on structural MRI and focused on binary classification. Tosun et al. (2014) [10] combined T1-weighted MRI with arterial spin labeling, a perfusion imaging technique, to predict amyloid positron emission tomography (PET) status using partial least squares regression. Although they included functional information, their model remained limited to MRI-based modalities and addressed classification rather than the estimation of CSF biomarkers. More recently, Lew et al. (2023) [11] applied deep learning to predict PET-defined amyloid, tau, and neurodegeneration (ATN) status from MRI and clinical data. However, their imaging input was limited to MRI, and they used PET-derived binary labels without estimating continuous CSF levels.

In contrast, our study introduces a deep learning framework that directly estimates CSF A β 42, A β 40, and the A β 42/A β 40 ratio as continuous variables using both T1-weighted MRI and cerebral blood flow SPECT. By formulating the task as a multitask regression problem and incorporating demographic variables, our model enables non-invasive, quantitative biomarker estimation. This approach extends beyond prior studies by integrating structural and functional imaging inputs and by targeting continuous CSF biomarker values rather than binary classification labels.

3. Preliminaries

3.1 Subjects and Clinical Data

We retrospectively collected clinical and imaging data from Hyogo Prefectural Harima-Himeji General Medical Center (formerly Hyogo Brain and Heart Center) between January 2014 and March 2024. Among 243 patients who underwent CSF analysis, 186 had both T1-weighted MRI and SPECT scans available. After excluding 33 cases due to incompatible MRI acquisition parameters, the final study cohort consisted of 153 patients.

The cohort included 63 male and 90 female patients, with a mean age of 70.9 years (standard deviation = 11.8; range: 37-90). Age and sex were incorporated as auxiliary variables in our prediction model.

3.2 Imaging Acquisition

T1-weighted MRI scans were acquired using Philips Achieva, Intera, or Ingenia scanners, with an in-plane resolution of 256×256 pixels, a slice thickness of 1.2 mm, and a field of view of 256 mm. Brain perfusion SPECT images were obtained using Siemens IP2 or GE Tandem 870 DR systems, with a matrix size of 128×128 , a slice thickness of 3.9 mm, and a field of view of 498.6 mm. All images were spatially normalized to a common reference space using SPM12 prior to being input into the model.

3.3 Biomarkers

Three CSF biomarkers were used as prediction targets: A β 42 and A β 40 (both measured in pg/mL), and the A β 42/40 ratio. A β 42 is recognized as a core pathological marker for AD, while A β 40 serves as a complementary measure that accounts for individual differences in amyloid production. The A β 42/40 ratio has been reported to offer higher diagnostic stability and specificity than A β 42 alone [12].

4. Proposed Method

This study proposes a 3D deep learning framework for estimating continuous CSF biomarker values, including A β 42, A β 40, and the A β 42/40 ratio, from multimodal neuroimaging data. The framework consists of two main components: the integration of structural and functional imaging modalities, and multitask regression modeling.

To preserve the complementary pathological significance of A β 42 and A β 40, as well as the diagnostic robustness of their ratio, all three biomarkers are predicted simultaneously. This formulation enables the use of multitask learning, which facilitates shared representation across related outputs and improves generalization.

4.1 Structural and Functional Image Integration

To effectively leverage both anatomical and functional brain information, we integrate T1-weighted MRI and brain perfusion SPECT into a single, spatially normalized multimodal input tensor. This process involves spatial alignment, tissue segmentation, nonlinear normalization, and intensity scaling to prepare the data for input into the deep learning model.

All image preprocessing is performed using SPM12 (Wellcome Trust Center for Neuroimaging, UCL) in MATLAB R2024a. For each subject, SPECT images are rigidly co-registered to the corresponding T1-weighted MRI using mutual information as the cost function. T1-weighted MRIs are then segmented into gray matter (GM) and white matter (WM) probability maps using the SPM unified segmentation algorithm.

To achieve accurate inter-subject normalization, we use DARTEL (Diffeomorphic Anatomical Registration Through Exponentiated Lie Algebra), a high-dimensional nonlinear registration method that iteratively estimates shape differences in brain anatomy. Individual deformation fields are generated and applied to spatially normalize

GM, WM, and SPECT images to Montreal Neurological Institute (MNI) space with 1.5 mm isotropic resolution. GM and WM maps are modulated using the Jacobian determinants of the deformation fields to preserve local volumetric information. In contrast, modulation is not applied to SPECT, since its voxel intensities reflect regional cerebral blood flow rather than structural volume. Applying structure-based modulation to functional signals could compromise their physiological interpretability and introduce confounding effects.

Although data are acquired using two different SPECT systems (Siemens IP2 or GE Tandem 870 DR systems), and imaging parameters vary between MRI (in-plane resolution of 256×256 pixels, a slice thickness of 1.2 mm, and a field of view of 256 mm) and SPECT (matrix size of 128×128 , a slice thickness of 3.9 mm, and a field of view of 498.6 mm) acquisitions as detailed in Section 3.2, all images are spatially normalized to MNI space with 1.5 mm isotropic voxels using DARTEL to unify anatomical locations and effectively standardize voxel dimensions across all modalities. While spatial normalization to MNI space unifies anatomical locations, it does not inherently harmonize absolute intensity scales across different scanners. We therefore rely on the model's capacity to integrate these differently processed modalities by learning from relative regional perfusion patterns, which are less susceptible to scanner-dependent intensity variations.

All GM, WM, and SPECT images are smoothed with a Gaussian kernel of 12 mm full width at half maximum (FWHM) to reduce inter-subject anatomical variability. The resulting GM, WM, and SPECT volumes are resampled to a uniform size of $128 \times 128 \times 128$ voxels and concatenated along the channel axis to form a three-channel 3D tensor. Each volume is independently scaled to $[0, 1]$ using min-max normalization. CSF biomarker values are transformed using a quantile transformer (scikit-learn v1.5.2) fitted on the training data, and subsequently scaled to $[0, 1]$ to facilitate network training.

4.2 Multitask Regression Model

We implement a multitask regression model based on a 3D convolutional neural network (CNN) to simultaneously estimate CSF A β 42, A β 40, and the A β 42/40 ratio from a multimodal input tensor and demographic variables. The overall structure of the model is illustrated in Fig. 1.

The input tensor, shaped $[3, 128, 128, 128]$, is passed through a $1 \times 1 \times 1$ convolutional layer with ReLU activation to adjust the channel dimensionality. This is followed by a shared feature extraction backbone based on the 3D ResNet-34 architecture. The network consists of four residual stages with increasing filter sizes of 64, 128, 256, and 512. Each stage contains multiple residual convolutional blocks with batch normalization and ReLU activations.

Demographic data (age and gender) are incorporated as auxiliary inputs. Gender is encoded as 0 for male and 1 for female, and both variables are scaled to $[0, 1]$ using min-max normalization. After flattening the image features extracted by the ResNet encoder, the demographic feature vector is concatenated to form a joint representation, which is passed through a fully connected layer with ReLU activation.

The shared representation is then fed into three independent output heads, each consisting of a fully connected layer that predicts one CSF biomarker. The outputs are scaled to $[0, 1]$ using the same quantile transformer applied during training, and inverse-transformed during evaluation to recover biomarker values in physical units.

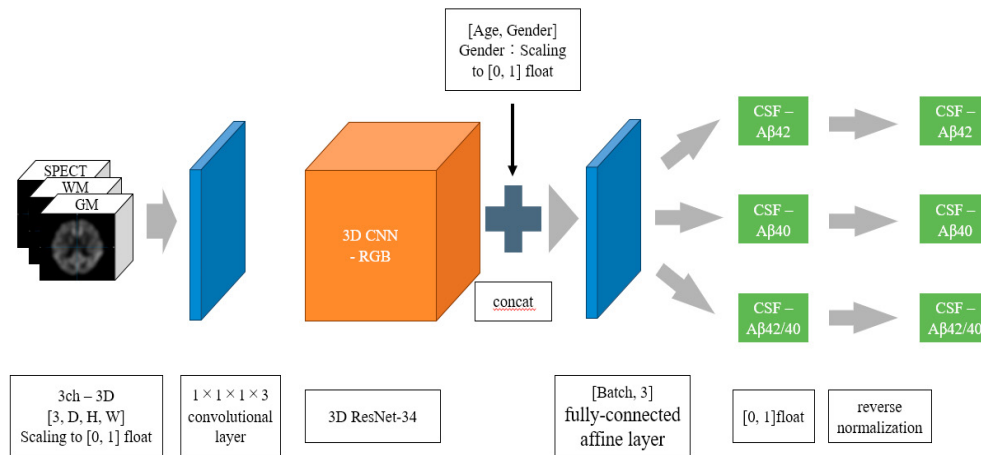


Fig. 1. Overview of the proposed image-to-biomarker estimation framework.

5. Experimental results

5.1 Experimental Setup

All experiments were conducted using Python v3.12 and PyTorch 2.4.0 on a single NVIDIA RTX A6000 GPU with 49GB GDDR6 memory. Model training used a weighted mean squared error (MSE) loss function, with fixed task weights of 1.2 for Aβ42, 0.7 for Aβ40, and 1.7 for the Aβ42/40 ratio. The model was trained with a batch size of 8 for 100 epochs. Optimization was performed using the AdamW optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and weight decay of $1e-3$. The initial learning rate was set to $1e-4$ and scheduled using cosine annealing with a warm-up phase of 10 epochs. Five-fold cross-validation was performed, stratified by a fixed random seed (42).

To assess the contribution of individual inputs, ablation studies were conducted by removing the SPECT modality and/or the age variable.

5.2 Evaluation metrics

We evaluated the model's performance using mean absolute error (MAE), root mean squared error (RMSE), and Pearson correlation coefficient (r). Each metric was computed separately for each biomarker and averaged over the cross-validation folds.

5.3 Results under different input configurations

The proposed model was evaluated under three input configurations:

- (1) Full Inputs (Age + Gender + SPECT + structural MRI)
- (2) No Age (Gender + SPECT + structural MRI)
- (3) No SPECT (Age + Gender + structural MRI)

Table 1 summarizes the mean performance values averaged over five cross-validation folds.

The best performance for A β 40 and A β 42/40 was achieved when both age and gender were included, along with SPECT images. Excluding age resulted in moderate performance degradation, while removing SPECT entirely led to the worst performance. These findings suggest that demographic features, particularly age, and functional imaging both contribute significantly to accurate estimation of A β 40 and A β 42/40.

In contrast, A β 42 exhibited a different trend. While the No Age configuration showed the lowest MAE and RMSE, the highest correlation ($r = 0.558$) was achieved when both age and gender were used. The model without SPECT yielded the lowest correlation ($r = -0.158$), indicating poor predictive alignment with ground truth. This suggests that the No Age configuration may yield predictions biased toward the population mean, failing to capture inter-individual variability, which SPECT helps recover.

Table 1. Comparison of prediction performance under different input configurations. Metrics include mean absolute error (MAE), root mean squared error (RMSE), and Pearson correlation coefficient (r). All values are averaged over five cross-validation folds.

	A β 42			A β 40			A β 42/40		
	MAE (pg/mL)	RMSE (pg/mL)	r	MAE (pg/mL)	RMSE (pg/mL)	r	MAE	RMSE	r
(1) Full Inputs	353	500	0.558	2933	3838	0.568	0.0292	0.0355	0.428
(2) No Age	320	437	0.531	3236	4238	0.492	0.0322	0.0395	0.285
(3) No SPECT	318	416	-0.158	3706	4588	0.133	0.0308	0.0381	0.150

5.4 Visualization and Interpretation

Scatter plots of predicted versus true values for each configuration are shown in Figs. 2–4. When both age and gender were included, predictions exhibited broad variance and moderate correlation. The No Age model (Fig. 3) also preserved the prediction range but showed slightly reduced correlation. In contrast, the model without SPECT (Fig. 4) produced tightly clustered predictions around the mean, failing to capture the spread of true values. This reinforces the importance of functional information in capturing individual biomarker variation.

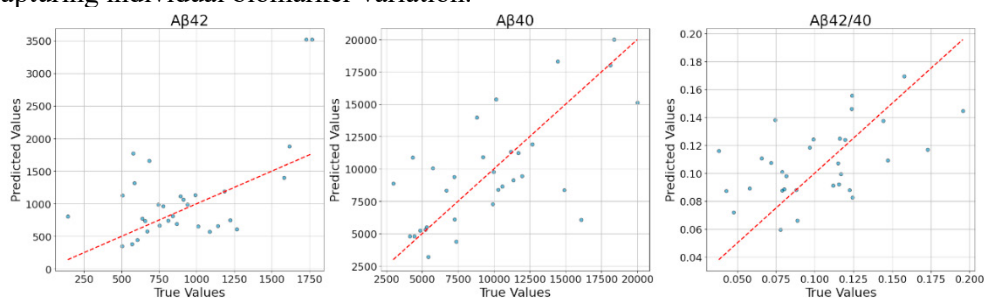


Fig. 2. True vs. predicted values (Full Inputs: Age + Gender + SPECT + structural MRI).

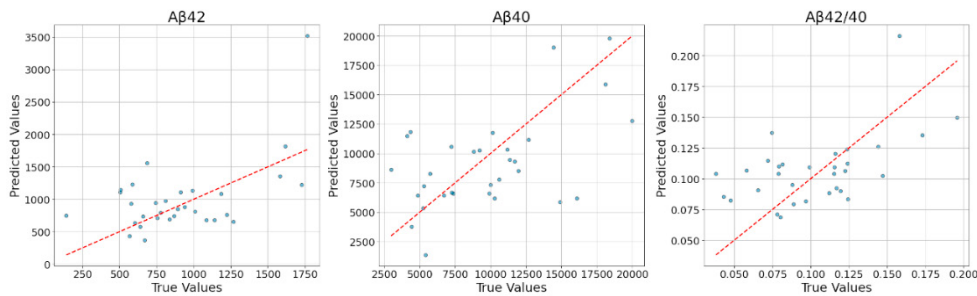


Fig. 3. True vs. predicted values (No Age: Gender + SPECT + structural MRI).

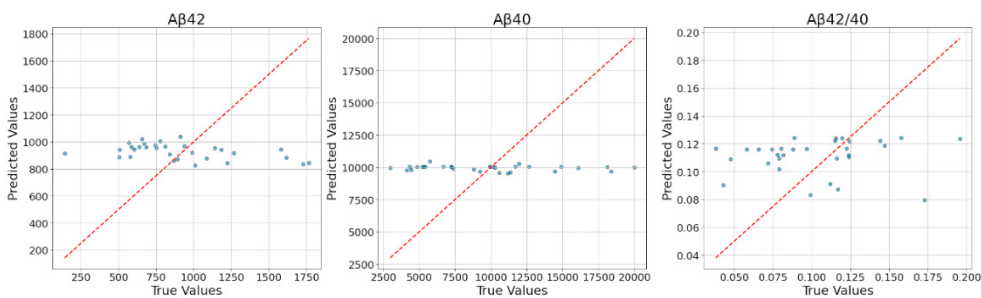


Fig. 4. True vs. predicted values (No SPECT: Age + Gender + structural MRI).

6. Discussion

In this study, we investigated whether integrating functional SPECT imaging with structural MRI improves CSF biomarker prediction compared to using MRI alone. The results clearly support this hypothesis, especially for A β 42.

As shown in Table 1, removing SPECT led to a marked decline in prediction performance for A β 42, with the correlation dropping from 0.558 (age + gender) to 0.531 (gender only), and further to -0.158 (no SPECT). The scatter plots confirmed this trend, revealing that SPECT provided critical functional information for capturing inter-individual variation in biomarker levels.

However, despite these improvements, the obtained RMSE of 500 pg/mL for A β 42, when viewed in absolute terms, represents a significant proportion of the established CSF A β 42 reference value (>540 pg/mL) [13]. For instance, if the true value is close to 550 pg/mL, an error of 500 pg/mL could yield predictions ranging from 50 to 1050 pg/mL, potentially leading to substantially altered clinical interpretations. This level of error suggests that our current prediction accuracy may not yet provide diagnostic precision comparable to direct CSF measurement via lumbar puncture. Although our model performs regression, directly assessing sensitivity and specificity (typically used for classification) is not straightforward. Nevertheless, such an error highlights the need for further improvement in sensitivity and specificity of early AD diagnosis when clinical decisions rely solely on our prediction model. Compared to other existing studies on similar prediction tasks, our model's RMSE still suggests room for

improvement, indicating that further improvement in accuracy is needed for practical clinical application.

Age also contributed positively to model performance, particularly for A β 40 and A β 42/40. When age was excluded, moderate but consistent increases in MAE and RMSE were observed, along with reduced correlation. However, the impact of removing age was much smaller compared to removing SPECT, indicating that age serves as a useful but non-substitutive predictor that complements the structural information.

A notable finding is the discrepancy between error metrics and correlation in A β 42 prediction under the No Age condition. Although the No Age model achieved the lowest MAE (320 pg/mL) and RMSE (437 pg/mL), its correlation ($r = 0.531$) was still lower than that of the age + gender condition ($r = 0.558$), and significantly higher than the no-SPECT case ($r = -0.158$). This suggests that low error values alone may be misleading, as they can result from predictions clustering around the mean without reflecting actual data variability.

Our results demonstrate that both SPECT and age are essential input features, with SPECT being particularly crucial for variance-sensitive targets like A β 42. Age contributes to prediction stability and complements structural information. The integration of these modalities leads to more robust and interpretable CSF biomarker prediction. Future work may benefit from incorporating additional clinical variables, such as APOE genotype, cognitive scores, or clinical staging measures, testing longitudinal consistency, and validating across external datasets to enhance generalizability and clinical applicability.

7. Future Work

Despite the promising results, several challenges remain for the clinical application of our proposed framework.

The observed RMSE of 500 pg/mL for A β 42, compared to its clinical reference value of over 540 pg/mL, suggests that although the model is promising, it still requires substantial improvements in accuracy. Since the model performs regression, it is not straightforward to assess diagnostic performance using sensitivity and specificity, which are typically applied in classification tasks. Future work will therefore include evaluation of these metrics by converting predicted values into categorical decisions based on established clinical cutoffs, in order to more clearly demonstrate the model's diagnostic utility.

In addition to its non-invasive nature, our method should be assessed for cost-effectiveness, particularly in comparison with existing invasive procedures such as lumbar puncture and expensive alternatives like amyloid PET. This will be important for determining its practical value in routine clinical settings.

Improving the explainability of the model is also a key goal. Identifying the specific brain regions or image-derived features that contribute most to the predictions will help build clinical trust and support informed decision-making.

Finally, as this study was conducted using data from a single Japanese institution, validation on external datasets that include diverse ethnic and geographic populations will be essential for confirming generalizability and promoting broader clinical applicability.

8. Conclusion

This study proposed a deep learning framework for non-invasive prediction of CSF A β biomarkers by integrating structural MRI, brain perfusion SPECT, and demographic data. The results clearly addressed our research question: integrating SPECT with MRI significantly improved the accuracy of biomarker estimation compared to using MRI alone, particularly for variance-sensitive targets such as A β 42. Brain perfusion SPECT captured inter-individual variability that MRI alone could not, while age contributed to prediction stability. These findings underscore the effectiveness of multimodal fusion for accurate and interpretable biomarker estimation, and suggest that incorporating additional clinical variables could further enhance model performance and clinical utility. For future work, key priorities include improving predictive accuracy, evaluating cost-effectiveness, enhancing model explainability, and validating generalizability across diverse external datasets.

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